FLIPKART LOGISTICS

OPTIMIZATION FOR DELIVERY ROUTES

SQL-driven analytics to reduce delays, optimize routes & improve agent efficiency

Dataset: 300 Orders | 20 Routes | 10 Warehouses | 50 Agents | 999 Checkpoints

Tool: MySQL Workbench | **Tasks:** Data Cleaning · Delay Analysis · Route Optimization · Warehouse · Agent · KPI Reporting

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PROJECT OVERVIEW

Challenge

Rising order volumes during festive seasons cause delays, route inefficiencies, and traffic disruptions affecting customer satisfaction.

Approach

Leverage 5 relational tables (Orders, Routes, Warehouses, Agents, Tracking) with SQL queries, CTEs, window functions and aggregations.

Objective

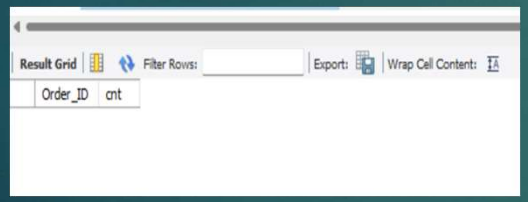
Build a SQL-driven analytics system to analyze delays, optimize routes, and enhance shipment & agent efficiency using data-driven insights.

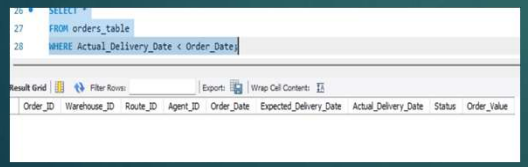
Outcome

Actionable recommendations on route optimization, warehouse bottlenecks, agent training and KPI dashboards for Ekart Logistics.

Data Cleaning & Preparation

- No duplicate records
- Converted data types
- Date format already standardized
- No null values in Traffic_Delay_Min
- No records found where Actual_Delivery_Date is before Order_Date





QUERIES USED

1. SELECT Order_ID, COUNT(*) AS cnt
FROM orders_table
GROUP BY Order_ID
HAVING COUNT(*) > 1;

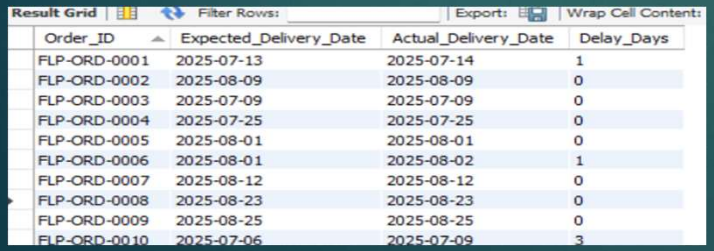
2. ALTER TABLE orders_table
MODIFY Order_Value DECIMAL(10,2);

3. SELECT * FROM orders_table
WHERE Actual_Delivery_Date < Order_Date;

Task 2:Delivery Delay Calculation

Query-

```
SELECT Order_ID,
  Expected_Delivery_Date,
   Actual_Delivery_Date,
  DATEDIFF(Actual_Delivery_Date,
Expected_Delivery_Date) AS Delay_Days
FROM orders_table;
```


Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Order_ID	Expected_Delivery_Date	Actual_Delivery_Date	Delay_Days
FLP-ORD-0001	2025-07-13	2025-07-14	1
FLP-ORD-0002	2025-08-09	2025-08-09	0
FLP-ORD-0003	2025-07-09	2025-07-09	0
FLP-ORD-0004	2025-07-25	2025-07-25	0
FLP-ORD-0005	2025-08-01	2025-08-01	0
FLP-ORD-0006	2025-08-01	2025-08-02	1
FLP-ORD-0007	2025-08-12	2025-08-12	0
FLP-ORD-0008	2025-08-23	2025-08-23	0
FLP-ORD-0009	2025-08-25	2025-08-25	0
FLP-ORD-0010	2025-07-06	2025-07-09	3

Order_ID	Expected_Delivery_Date	Actual_Delivery_Date	Delay_Days
FLP-ORD-0001	2025-07-13	2025-07-14	1
FLP-ORD-0002	2025-08-09	2025-08-09	0
FLP-ORD-0003	2025-07-09	2025-07-09	0
FLP-ORD-0004	2025-07-25	2025-07-25	0
FLP-ORD-0005	2025-08-01	2025-08-01	0
FLP-ORD-0006	2025-08-01	2025-08-02	1
FLP-ORD-0007	2025-08-12	2025-08-12	0
FLP-ORD-0008	2025-08-23	2025-08-23	0

Top Delayed Orders

Query-

```
SELECT  Order_ID,
 DATEDIFF(Actual_Delivery_Date, Expected_Delivery_Date) AS Delay_Days
FROM orders_table
ORDER BY Delay_Days DESC
LIMIT 10;
```

Order_ID	Delays_days
FLP-ORD-0166	3
FLP-ORD-0118	3
FLP-ORD-0025	3
FLP-ORD-0079	3
FLP-ORD-0092	3
FLP-ORD-0077	3
FLP-ORD-0129	3
FLP-ORD-0165	3
FLP-ORD-0010	3
FLP-ORD-0121	3

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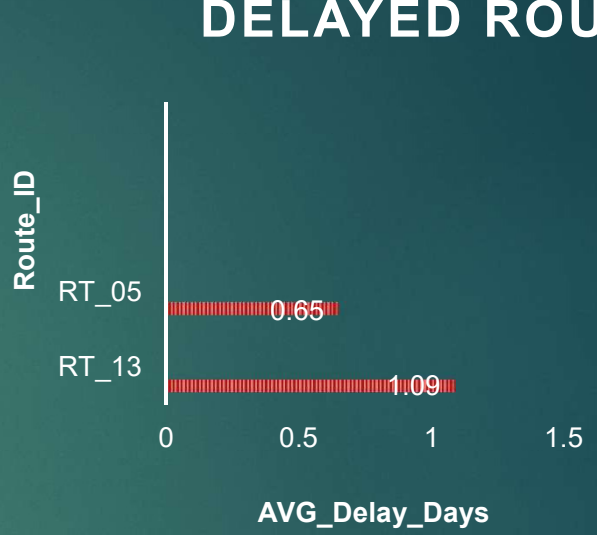
Top Delayed Routes

SELECT Route_ID, AVG(DATEDIFF(Actual_Delivery_Date, Expected_Delivery_Date)) AS Avg_Delay_Days FROM orders_table GROUP BY Route_ID ORDER BY Avg_Delay_Days DESC LIMIT 10;

Result Grid | Filter Rows:

Route_ID	Avg_Delay_Days
RT_13	1.0909
RT_05	0.6500
RT_14	0.6471
RT_09	0.6250
RT_02	0.5882
RT_18	0.5714
RT_01	0.5385
RT_12	0.5000
RT_06	0.5000
RT_16	0.5000



KEY INSIGHTS:
Route RT_13 was identified as the most delayed route with an average delay of 1.09 days.
Other routes such as RT_05, RT_14, and RT_09 also showed moderate delays. This indicates that certain routes require operational improvements to reduce delivery time.
These routes are inefficient and need optimization

-Targeted improvements in high-delay routes can significantly enhance overall delivery efficiency

- Higher Avg Delay = Poor route planning
- Lower Avg Delay = Efficient route

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Warehouse Performance Analysis

```
SELECT Warehouse_ID,Order_ID,DATEDIFF(Actual_Delivery_Date,
Expected_Delivery_Date) AS Delay_Days,    RANK() OVER (
PARTITION BY Warehouse_ID        ORDER BY
DATEDIFF(Actual_Delivery_Date, Expected_Delivery_Date) DESC)
AS Rank_in_Warehouse FROM orders_table;
```

Warehouse	Order_ID	Delay	Rank
WH_01	FLP-ORD-0077	3	1
WH_01	FLP-ORD-0194	3	1
WH_01	FLP-ORD-0210	2	3
WH_01	FLP-ORD-0055	2	3
WH_01	FLP-ORD-0148	1	5
WH_01	FLP-ORD-0296	1	5
WH_01	FLP-ORD-0002	0	7

Lowest delay = best warehouse

Orders ranked by delay within each warehouse

Key Insights:

- Majority deliveries are on time (0 delay)
- Maximum delay observed: 3 days
- Certain routes (RT_13) show higher delays
- Warehouse performance varies
- Improvement needed in specific routes like RT_13.

Conclusion:-

- Overall system performs well
- Delays are limited but concentrated
- Targeted improvements can enhance efficiency

Task 3: Route Optimization Insights

- Route optimization is the process of finding the most efficient path or sequence of stops between multiple locations.
- The goal is to minimize the time, distance, cost or fuel use.

Average Delivery Time Per Route

```
SELECT
Route_ID,
ROUND(AVG(DATEDIFF(Actual_Delivery_Date, Order_Date)),2) AS Avg_Delivery_Time
FROM orders_table
GROUP BY Route_ID;
```

Result Grid | Filter Rows:

Route_ID	Avg_Delivery_Time
RT_20	4.80
RT_14	4.88
RT_12	4.61
RT_10	4.42
RT_05	4.90
RT_19	4.07
RT_17	3.80
RT_11	4.00
RT_15	5.06
RT_13	4.73
RT_08	4.50

RT_15 takes the longest with 5.06 days on average, followed by RT_14 at 4.88 days and RT_05 at 4.90 days. RT_17 is the fastest with only 3.80 days.

Average Traffic Delay

- SELECT Route_ID,AVG(Traffic_Delay_Min) AS Avg_Traffic_Delay FROM routes_table GROUP BY route_id;

Route_ID	Avg_Traffic_Delay
RT_01	67.0000
RT_02	30.0000
RT_03	29.0000
RT_04	23.0000
RT_05	55.0000
RT_06	15.0000
RT_07	58.0000
RT_08	90.0000
RT_09	83.0000
RT_10	15.0000
RT_11	20.0000
RT_12	58.0000
RT_13	56.0000
RT_14	36.0000

RT_08 has the highest traffic delay at 90 minutes, followed by RT_09 at 83 minutes and RT_17 at 81 minutes. On the other end, RT_06 and RT_10 have the lowest traffic delay at just 15 minutes

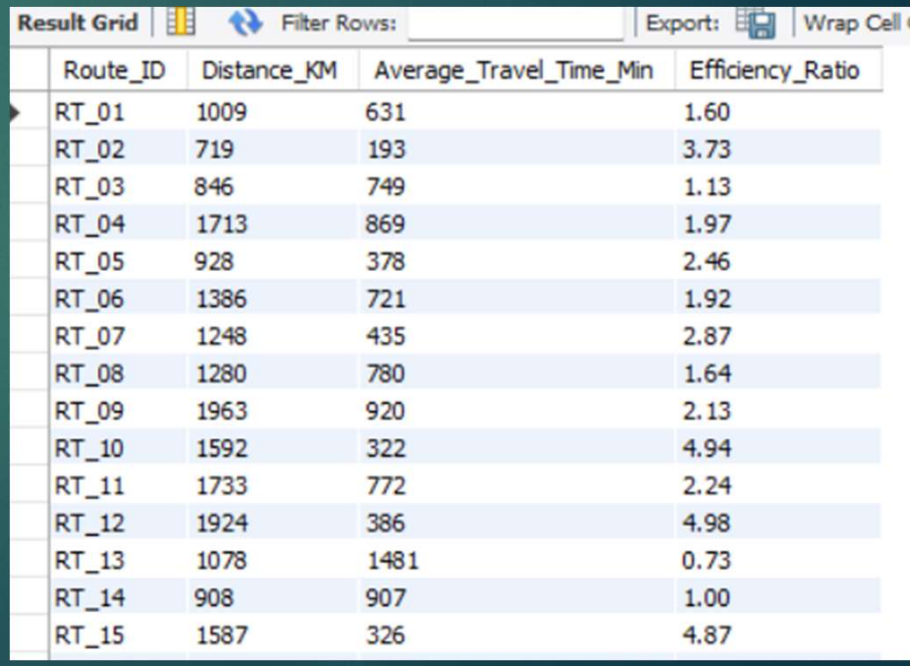
Efficiency Ratio

- SELECT r.Route_ID,r.Distance_KM,r.Average_Travel_Time_Min,ROUND(r.Distance_KM / r.Average_Travel_Time_Min, 2) AS Efficiency_RatioFROM routes_table r;

A higher ratio means the route covers more distance per minute of travel, which means it is more efficient

RT_12 has the highest ratio of 4.98 and RT_10 has 4.94 — these are the most efficient routes.

But RT_13 has the lowest ratio of just 0.73 — meaning it covers very little distance relative to its travel time. RT_14 is also poor at 1.00. These are the routes wasting the most time and need to be restructured or rerouted.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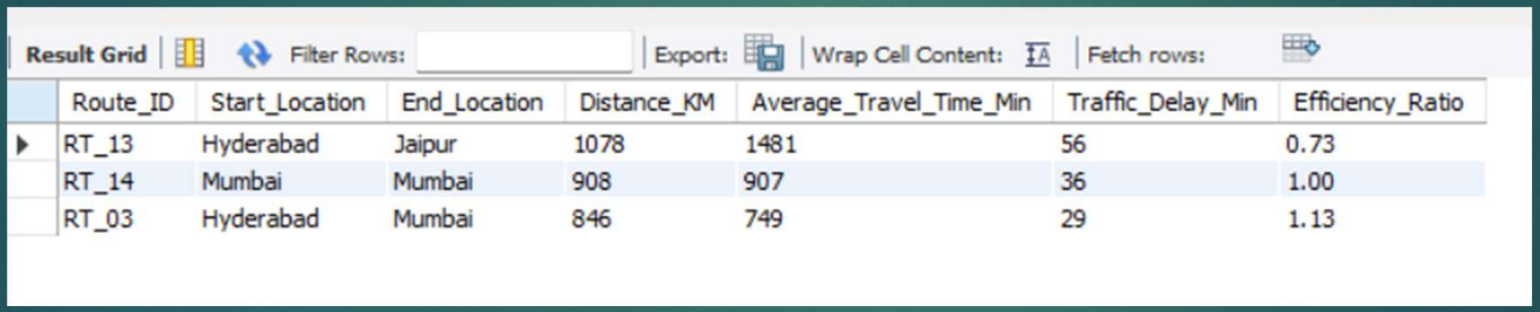
Route_ID	Distance_KM	Average_Travel_Time_Min	Efficiency_Ratio
RT_01	1009	631	1.60
RT_02	719	193	3.73
RT_03	846	749	1.13
RT_04	1713	869	1.97
RT_05	928	378	2.46
RT_06	1386	721	1.92
RT_07	1248	435	2.87
RT_08	1280	780	1.64
RT_09	1963	920	2.13
RT_10	1592	322	4.94
RT_11	1733	772	2.24
RT_12	1924	386	4.98
RT_13	1078	1481	0.73
RT_14	908	907	1.00
RT_15	1587	326	4.87

Worst 3 Routes (Efficiency)

- SELECT Route_ID,Start_Location,End_Location,Distance_KM,Average_Travel_Time_Min,Traffic_Delay_Min,ROUND(Distance_KM / Average_Travel_Time_Min, 2) AS Efficiency_Ratio FROM routes_table ORDER BY Efficiency_Ratio ASC LIMIT 3;

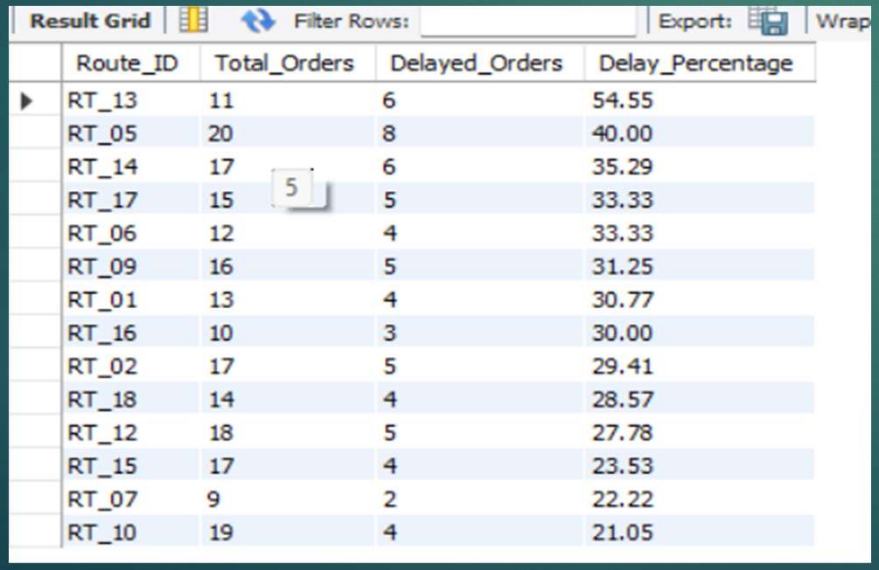


Route_ID	Start_Location	End_Location	Distance_KM	Average_Travel_Time_Min	Traffic_Delay_Min	Efficiency_Ratio
RT_13	Hyderabad	Jaipur	1078	1481	56	0.73
RT_14	Mumbai	Mumbai	908	907	36	1.00
RT_03	Hyderabad	Mumbai	846	749	29	1.13

Lower ratio = inefficient routes

Routes with >20% Delayed Shipments

- SELECT Route_ID, COUNT(*) AS Total_Orders, SUM(CASE WHEN Actual_Delivery_Date > Expected_Delivery_Date THEN 1 ELSE 0 END) AS Delayed_Orders, ROUND(SUM(CASE WHEN Actual_Delivery_Date > Expected_Delivery_Date THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS Delay_Percentage FROM orders_table GROUP BY Route_ID HAVING SUM(CASE WHEN Actual_Delivery_Date > Expected_Delivery_Date THEN 1 ELSE 0 END) * 100.0 / COUNT(*) > 20 ORDER BY Delay_Percentage DESC;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Route_ID	Total_Orders	Delayed_Orders	Delay_Percentage
RT_13	11	6	54.55
RT_05	20	8	40.00
RT_14	17	6	35.29
RT_17	15	5	33.33
RT_06	12	4	33.33
RT_09	16	5	31.25
RT_01	13	4	30.77
RT_16	10	3	30.00
RT_02	17	5	29.41
RT_18	14	4	28.57
RT_12	18	5	27.78
RT_15	17	4	23.53
RT_07	9	2	22.22
RT_10	19	4	21.05

High-Risk Routes (Immediate Attention)
-These routes have very high delay %:
RT_13 → 54.55% (worst)
RT_05 → 40.00%
RT_14 → 35.29%
More than 1 out of 3 deliveries are getting delayed.
-RT_13 appears everywhere (Worst delay + Worst %).

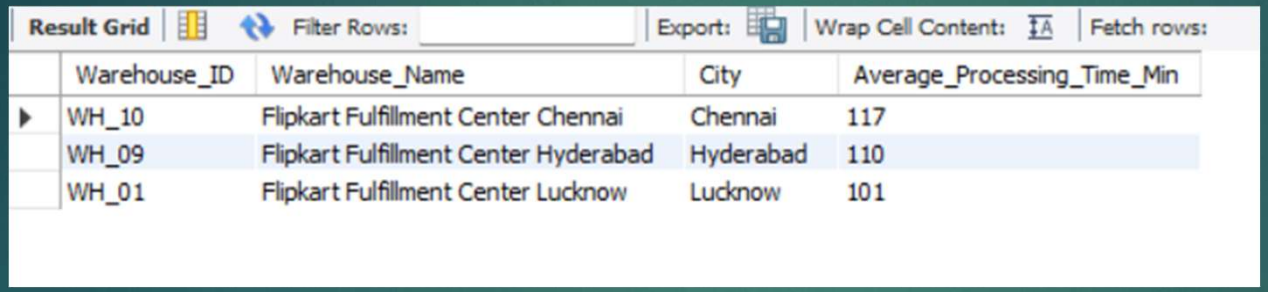
FINAL RECOMMENDATIONS

- Fix High-Risk Routes First:
- Focus on:
- RT_13
- RT_05
- RT_14
- Review route planning
- Check traffic congestion patterns
- Reassign delivery agents if needed
- Investigate Root Cause
- Traffic issues

Improve Scheduling
Use Best Routes as Benchmark

TASK:4~Warehouse Performance Analysis

- Top 3 Warehouses with Highest Processing Time



Warehouse_ID	Warehouse_Name	City	Average_Processing_Time_Min
WH_10	Flipkart Fulfillment Center Chennai	Chennai	117
WH_09	Flipkart Fulfillment Center Hyderabad	Hyderabad	110
WH_01	Flipkart Fulfillment Center Lucknow	Lucknow	101

- SELECT Warehouse_ID, Warehouse_Name, City, Average_Processing_Time_Min FROM warehouse_table ORDER BY Average_Processing_Time_Min DESC LIMIT 3;
- WH_10 Chennai has the highest processing time at 117 minutes, followed by WH_09 Hyderabad (110 min) and WH_01 Lucknow (101 min). These 3 warehouses need operational improvements to reduce processing delays.

Total vs Delayed Shipments per Warehouse

- SELECT Warehouse_ID,COUNT(*) AS Total_Orders,SUM(CASE WHEN Actual_Delivery_Date > Expected_Delivery_Date THEN 1 ELSE 0 END) AS Delayed_Orders FROM orders_table GROUP BY Warehouse_ID;
- **Worst Performing Warehouses**

------WH_10 → 11 delayed orders (highest)

------WH_08 → 10 delayed orders

------WH_02 & WH_03 → 9 delayed orders

- These warehouses are major contributors to delivery delays and require operational improvements
- Best Performing Warehouse
- WH_09 → 43 total orders but only 7 delayed

Warehouse_ID	Total_Orders	Delayed_Orders
WH_07	26	8
WH_01	24	6
WH_04	31	8
WH_02	27	9
WH_09	43	7
WH_10	30	11
WH_08	38	10
WH_05	29	8
WH_06	27	6
WH_03	25	9

Bottleneck warehouses (CTE)

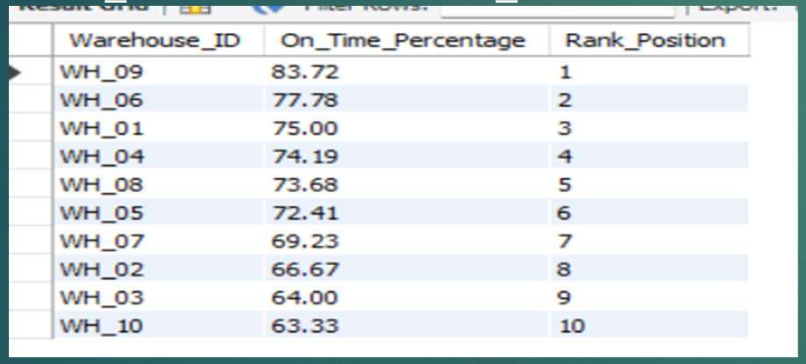
- WITH avg_time AS (SELECT AVG(DATEDIFF(Actual_Delivery_Date, Order_Date)) AS Global_Avg FROM orders_table)SELECT o.Warehouse_ID,AVG(DATEDIFF(o.Actual_Delivery_Date, o.Order_Date)) AS Avg_Time FROM orders_table o CROSS JOIN avg_time a GROUP BY o.Warehouse_ID HAVING AVG(DATEDIFF(o.Actual_Delivery_Date, o.Order_Date)) > (SELECT Global_Avg FROM avg_time) ORDER BY Avg_Time DESC;

Warehouse_ID	Avg_Time
WH_02	5.2963
WH_10	4.8333
WH_06	4.7407
WH_05	4.6897

- **WH_02** is the biggest bottleneck (highest delay)
- Followed by **WH_10, WH_06, WH_05**
- These warehouses need:
- Process optimization
- Resource allocation
- Capacity improvement

Rank warehouses based on on-time delivery percentage

- SELECT Warehouse_ID,ROUND(SUM(CASE WHEN Actual_Delivery_Date <= Expected_Delivery_Date THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS On_Time_Percentage,RANK() OVER (ORDER BY SUM(CASE WHEN Actual_Delivery_Date <= Expected_Delivery_Date THEN 1 ELSE 0 END) * 1.0 / COUNT(*) DESC) AS Rank_Position FROM orders_table GROUP BY Warehouse_ID;



Warehouse_ID	On_Time_Percentage	Rank_Position
WH_09	83.72	1
WH_06	77.78	2
WH_01	75.00	3
WH_04	74.19	4
WH_08	73.68	5
WH_05	72.41	6
WH_07	69.23	7
WH_02	66.67	8
WH_03	64.00	9
WH_10	63.33	10

WH_09 =Best performing warehouse
WH_10 =Worst performer

KEY INSIGHTS (Warehouse Performance)

- WH_09 achieved the highest on-time delivery rate (83.72%)
- WH_10 showed the lowest performance (63.33%)
- Bottleneck warehouses like WH_02 and WH_10 require optimization
- Some warehouses (e.g WH_06) handle high processing time but still maintain good delivery performance
- Operational improvements are required in low-performing warehouses

Task 5: Delivery Agent Performance

- Rank agents (per route) by on-time delivery percentage
- SELECT Agent_ID,Agent_Name,Route_ID,Avg_Speed_KMPH,On_Time_Delivery_Percentage,Experience_Years,RANK() OVER (PARTITION BY Route_ID ORDER BY On_Time_Delivery_Percentage DESC) AS Rank_In_Route FROM deliveryagents_table ORDER BY Route_ID, Rank_In_Route;

Agent_ID	Agent_Name	Route_ID	Avg_Speed_KMPH	On_Time_Delivery_Percentage	Experience_Years	Rank_In_Route
AG_049	Kiran Reddy	RT_01	37.3	97.2	2.6	1
AG_047	Pooja Patel	RT_01	48.2	81.2	3.5	2
AG_002	Vikram Nair	RT_01	48.6	73.2	9	3
AG_038	Arun Reddy	RT_02	35.1	91.6	6.7	1
AG_007	Vikram Patel	RT_02	39.8	85.9	1.7	2
AG_020	Kiran Kumar	RT_02	48	85.1	1.1	3
AG_028	Priya Nair	RT_02	47.4	81.7	6.3	4
AG_026	Kiran Patel	RT_02	54.2	72.2	2.7	5
AG_013	Rajesh Patel	RT_03	53.1	94.1	1.6	1
AG_001	Arun Nair	RT_03	42.7	86.7	5.2	2
AG_017	Rajesh Gupta	RT_04	36.2	90.4	8.2	1
AG_050	Vikram Kumar	RT_04	46.9	79.7	8.1	2
AG_040	Anita Gupta	RT_05	45.9	93.4	6.2	1
AG_041	Amit Sharma	RT_05	35.8	80.2	9.4	2
AG_021	Anita Patel	RT_06	43.1	83.5	8.8	1

Agents with on-time % < 80%

- SELECT Agent_ID, Agent_Name,Route_ID, Avg_Speed_KMPH, On_Time_Delivery_Percentage,Experience_Years, CASE WHEN On_Time_Delivery_Percentage < 73 THEN 'Critical – Below 73%' WHEN On_Time_Delivery_Percentage < 77 THEN 'Poor – 73% to 77%' ELSE 'Below Average – 77% to 80%' END AS Performance_Level FROM deliveryagents_table WHERE On_Time_Delivery_Percentage < 80 ORDER BY On_Time_Delivery_Percentage ASC;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Agent_ID	Agent_Name	Route_ID	Avg_Speed_KMPH	On_Time_Delivery_Percentage	Experience_Years	Performance_Level
AG_006	Kiran Kumar	RT_20	41.5	70.5	9.4	Critical – Below 73%
AG_019	Vikram Sharma	RT_08	52.5	72.1	8.3	Critical – Below 73%
AG_026	Kiran Patel	RT_02	54.2	72.2	2.7	Critical – Below 73%
AG_004	Vikram Nair	RT_06	45.2	73	6.6	Poor – 73% to 77%
AG_035	Anita Patel	RT_09	37.3	73	6	Poor – 73% to 77%
AG_002	Vikram Nair	RT_01	48.6	73.2	9	Poor – 73% to 77%
AG_008	Meena Kumar	RT_18	50.3	73.6	5.3	Poor – 73% to 77%
AG_014	Pooja Reddy	RT_16	39.3	73.7	9.4	Poor – 73% to 77%
AG_012	Rajesh Nair	RT_18	39.6	76.2	1.6	Poor – 73% to 77%
AG_011	Sneha Sharma	RT_17	44	76.9	1.6	Poor – 73% to 77%
AG_036	Vikram Reddy	RT_20	39.2	79.6	7.2	Below Average – 7...
AG_050	Vikram Kumar	RT_04	46.9	79.7	8.1	Below Average – 7...
AG_032	Pooja Nair	RT_09	37.6	79.8	1.6	Below Average – 7...

avg speed of Top 5 vs Bottom 5 agents using subqueries

- SELECT 'Top 5 Agents (Best On-Time %)' AS Agent_Group, ROUND(AVG(Avg_Speed_KMPH), 2) AS Avg_Speed_KMPH FROM (SELECT Avg_Speed_KMPH FROM deliveryagents_table ORDER BY On_Time_Delivery_Percentage DESC LIMIT 5) top5 UNION ALL SELECT 'Bottom 5 Agents (Worst On-Time %)' AS Agent_Group, ROUND(AVG(Avg_Speed_KMPH), 2) AS Avg_Speed_KMPH FROM (SELECT Avg_Speed_KMPH FROM deliveryagents_table ORDER BY On_Time_Delivery_Percentage ASC LIMIT 5) bottom5;

Agent_Group	Avg_Speed_KMPH
Top 5 Agents (Best On-Time %)	44.08
Bottom 5 Agents (Worst On-Time %)	46.14

KEY INSIGHTS- Bottom agents drive **FASTER** (46.14 kmph) than top agents (44.08 kmph). Speed is NOT the problem — they are rushing deliveries and skipping proper handover steps. The solution is accuracy training, not speed improvement

Suggestions for Low Performers

- **1. Agents who are Critical (below 73%) with high experience (5+ years)** → AG_006 (9.4 yrs), AG_019 (8.3 yrs), AG_004 (6.6 yrs), AG_035 (6.0 yrs), AG_014 (9.4 yrs) **Suggestion:** These agents are experienced but underperforming — reassign them to shorter or less congested routes (like RT_10, RT_11 which have 0% delay) instead of training.
- **2. Agents who are Critical with low experience (below 3 years)** → AG_026 (2.7 yrs) **Suggestion:** Enroll in a structured delivery training program and pair with a top-performing agent on the same route before working independently.
- **3. Agents driving too fast (above 48 kmph) but delivering poorly** → AG_019 (52.5 kmph), AG_026 (54.2 kmph), AG_008 (50.3 kmph), AG_002 (48.6 kmph) **Suggestion:** Speed is not the problem — accuracy is. Enforce a maximum speed policy and focus training on proper handover, customer confirmation, and proof-of-delivery steps.
- **4.General workload balancing**
- → RT_18 has 2 low performers (AG_008 and AG_012) on the same route
- Suggestion: Redistribute orders from overloaded routes to better-performing routes. Use the on-time % data monthly to rotate agents across routes and prevent burnout.

Task 6: Shipment Tracking Analytics

- For each order, the last checkpoint and time.
- WITH RankedCheckpoints AS (SELECT Order_ID,Checkpoint,Checkpoint_Time,Delay_Reason,Delay_Minutes,ROW_NUMBER() OVER (PARTITION BY Order_ID ORDER BY Checkpoint_Time DESC) AS rn FROM shipmenttracking_table)SELECT Order_ID, Checkpoint AS Last_Checkpoint, Checkpoint_Time AS Last_Checkpoint_Time,Delay_Reason,Delay_Minutes FROM RankedCheckpoints WHERE rn = 1 ORDER BY Order_ID;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

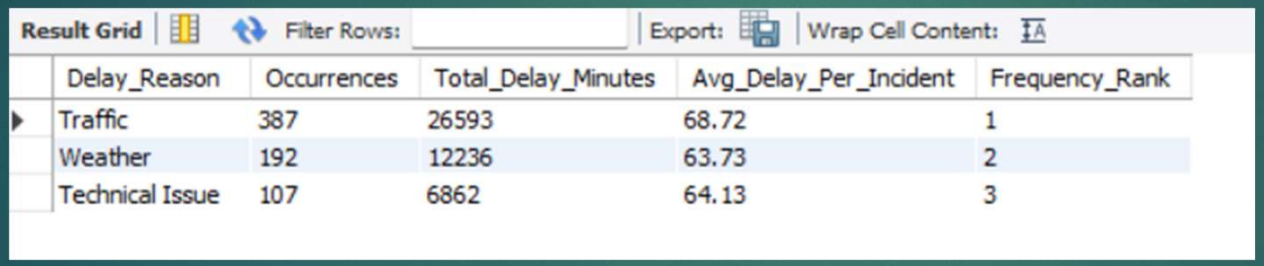
Order_ID	Last_Checkpoint	Last_Checkpoint_Time	Delay_Reason	Delay_Minutes
FLP-ORD-0002	Hub_4_Bengaluru	2025-08-24 10:27:00	None	0
FLP-ORD-0003	Hub_1_Mumbai	2025-08-28 16:54:00	None	0
FLP-ORD-0004	Hub_5_Jaipur	2025-08-24 23:15:00	None	0
FLP-ORD-0005	Hub_3_Pune	2025-08-22 11:50:00	None	0
FLP-ORD-0006	Hub_2_Lucknow	2025-08-13 05:48:00	Weather	67
FLP-ORD-0007	Hub_4_Hyderabad	2025-08-03 03:43:00	W Weather	38
FLP-ORD-0008	Hub_3_Ahmedabad	2025-07-13 00:43:00	Traffic	103
FLP-ORD-0009	Hub_2_Jaipur	2025-08-12 13:08:00	Traffic	92
FLP-ORD-0010	Hub_2_Bengaluru	2025-08-04 00:11:00	None	0
FLP-ORD-0011	Hub_1_Jaipur	2025-08-28 04:41:00	Traffic	112

Result 4

FLP-ORD-0002, the last checkpoint was Hub 4 Bengaluru on 24th August with no delay.
For FLP-ORD-0006, the last checkpoint was Hub 2 Lucknow with a Weather delay of 67 minutes.

The most common delay reasons

- SELECT Delay_Reason, COUNT(*) AS Occurrences, SUM(Delay_Minutes) AS Total_Delay_Minutes, ROUND(AVG(Delay_Minutes), 2) AS Avg_Delay_Per_Incident,RANK() OVER (ORDER BY COUNT(*) DESC) AS Frequency_Rank FROM shipmenttracking_table WHERE Delay_Reason IS NOT NULL AND UPPER(TRIM(Delay_Reason)) != 'NONE' GROUP BY Delay_Reason ORDER BY Occurrences DESC;

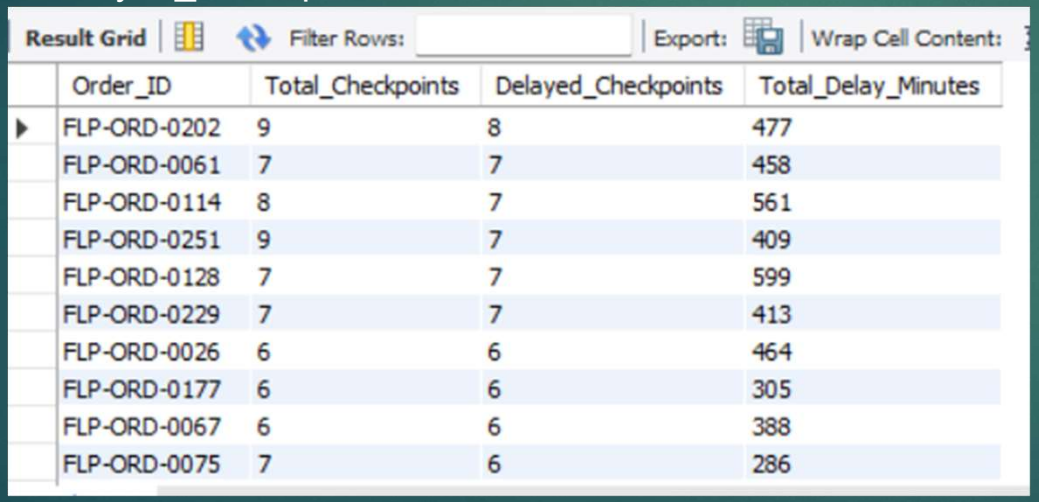

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Delay_Reason	Occurrences	Total_Delay_Minutes	Avg_Delay_Per_Incident	Frequency_Rank
Traffic	387	26593	68.72	1
Weather	192	12236	63.73	2
Technical Issue	107	6862	64.13	3

KEY INSIGHTS:- Traffic is the #1 cause of delays — 387 occurrences adding up to 26,593 total minutes of delay. That's 443 hours lost just due to traffic. Real-time route rerouting during peak hours would significantly reduce this.

orders with >2 delayed checkpoints

- SELECT Order_ID,COUNT(*) AS Total_Checkpoints, SUM(CASE WHEN Delay_Minutes > 0 THEN 1 ELSE 0 END) AS Delayed_Checkpoints,SUM(Delay_Minutes) AS Total_Delay_Minutes FROM shipmenttracking_table GROUP BY Order_ID HAVING SUM(CASE WHEN Delay_Minutes > 0 THEN 1 ELSE 0 END) > 2 ORDER BY Delayed_Checkpoints DESC;


Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Order_ID	Total_Checkpoints	Delayed_Checkpoints	Total_Delay_Minutes
FLP-ORD-0202	9	8	477
FLP-ORD-0061	7	7	458
FLP-ORD-0114	8	7	561
FLP-ORD-0251	9	7	409
FLP-ORD-0128	7	7	599
FLP-ORD-0229	7	7	413
FLP-ORD-0026	6	6	464
FLP-ORD-0177	6	6	305
FLP-ORD-0067	6	6	388
FLP-ORD-0075	7	6	286

Task 7: Advanced KPI Reporting

- Average Delivery Delay per Region (Start_Location)
- SELECT r.Start_Location AS Region,COUNT(o.Order_ID) AS Total_Orders,ROUND(AVG(DATEDIFF(o.Actual_Delivery_Date, o.Expected_Delivery_Date)),2) AS Avg_Delay_Days,SUM(CASE WHEN DATEDIFF(o.Actual_Delivery_Date,o.Expected_Delivery_Date) > 0 THEN 1 ELSE 0 END) AS Delayed_Orders,ROUND(100.0 * SUM(CASE WHEN DATEDIFF(o.Actual_Delivery_Date,o.Expected_Delivery_Date) <= 0 THEN 1 ELSE 0 END) / COUNT(o.Order_ID), 2) AS OnTime_Pct FROM orders_table o JOIN routes_table r ON o.Route_ID = r.Route_ID GROUP BY r.Start_Location ORDER BY Avg_Delay_Days DESC;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

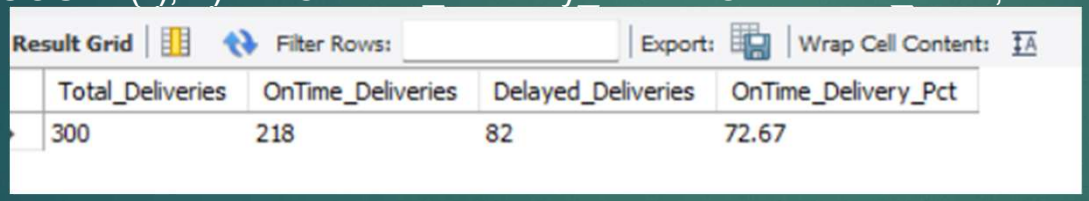
Region	Total_Orders	Avg_Delay_Days	Delayed_Orders	OnTime_Pct
Ahmedabad	33	0.61	10	69.70
Pune	36	0.56	11	69.44
Lucknow	13	0.54	4	69.23
Bengaluru	12	0.50	4	66.67
Mumbai	51	0.47	16	68.63
Hyderabad	155	0.38	37	76.13

Ahmedabad is the worst performing region with an average delay of 0.61 days and only 69.70% on-time rate, out of 33 orders. Pune is second worst at 0.56 days.

Hyderabad is the best performing region with just 0.38 days average delay and 76.13% on-time rate — and it also has the highest volume with 155 orders, which makes it even more impressive

On-Time Delivery %

- SELECT COUNT(*) AS Total_Deliveries,SUM(CASE WHEN DATEDIFF(Actual_Delivery_Date,Expected_Delivery_Date) <= 0 THEN 1 ELSE 0 END) AS OnTime_Deliveries,SUM(CASE WHEN DATEDIFF(Actual_Delivery_Date,Expected_Delivery_Date) > 0 THEN 1 ELSE 0 END) AS Delayed_Deliveries,ROUND(100.0 * SUM(CASE WHEN DATEDIFF(Actual_Delivery_Date,Expected_Delivery_Date) <= 0 THEN 1 ELSE 0 END) / COUNT(*), 2) AS OnTime_Delivery_Pct FROM orders_table;



Total_Deliveries	OnTime_Deliveries	Delayed_Deliveries	OnTime_Delivery_Pct
300	218	82	72.67

Average Traffic Delay per Route

- SELECT Route_ID, Start_Location, End_Location, Traffic_Delay_Min AS Avg_Traffic_Delay_Min, RANK() OVER (ORDER BY Traffic_Delay_Min DESC) AS Delay_Rank, CASE WHEN Traffic_Delay_Min >= 80 THEN 'High Traffic Impact' WHEN Traffic_Delay_Min >= 50 THEN 'Medium Traffic Impact' ELSE 'Low Traffic Impact' END AS Traffic_Category FROM routes_table ORDER BY Traffic_Delay_Min DESC;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Route_ID	Start_Location	End_Location	Avg_Traffic_Delay_Min	Delay_Rank	Traffic_Category
RT_08	Pune	Pune	90	1	High Traffic Impact
RT_15	Hyderabad	Jaipur	87	2	High Traffic Impact
RT_09	Ahmedabad	Mumbai	83	3	High Traffic Impact
RT_17	Mumbai	Lucknow	81	4	High Traffic Impact
RT_01	Lucknow	Bengaluru	67	5	Medium Traffic Impact
RT_07	Mumbai	Lucknow	58	6	Medium Traffic Impact
RT_12	Hyderabad	Lucknow	58	6	Medium Traffic Impact
RT_13	Hyderabad	Jaipur	56	8	Medium Traffic Impact
RT_05	Pune	Pune	55	9	Medium Traffic Impact
RT_19	Hyderabad	Lucknow	50	10	Medium Traffic Impact

RT_08 from Pune to Pune has the highest traffic delay at 90 minutes, ranked number 1. RT_15 from Hyderabad to Jaipur is second at 87 minutes. RT_09 and RT_17 also fall in the High Traffic Impact category at 83 and 81 minutes. Routes between 50 and 80 minutes are Medium Impact — and anything below 50 is Low Impact.

Conclusion

- Overall on-time rate: 72.67% (12% below 85% industry benchmark)
- RT_13 is the worst route — 54.55% delayed
- WH_10 Chennai is the biggest bottleneck — 117 min processing time
- Traffic = #1 delay cause — 387 incidents, 443 hours lost
- 13 agents (26%) need performance improvement
- Recommendation:~The main recommendation is to prioritize RT_13, reduce processing time in WH_10, implement real-time traffic rerouting, and train the 13 underperforming agents.